

24 — Ammonia Main + MGO Aux: an 11× factor gap on one vessel

Proves · Ammonia main + MGO aux: an 11× factor gap on one vessel, the strongest fuel guard in the suite.

Mechanics this scenario turns on

- ME and AE each use **their own fuel's** CO₂ factor (MGO/MDO 3.93267 · HFO 3.114 · LNG 2.753 · Ammonia 0.35154). The two per-engine cards must sum to the panel total.

Panels described below · Why this is the strongest fuel guard in the suite · Verified figures

Anything not described here — the mechanics above name the step that produced it; [00-ORIENTATION](#) Part 6 has the full number-to-step index.

Trust · characterisation snapshot, generated from the code. It detects change; it does NOT prove correctness. Figures marked *pending reference verification* have never been checked against anything outside the application.

Read after · scenario 13.

The widest CO₂ spread the model can produce, on scenario 19's plant so both engines burn.

Ammonia	0.35154 kg CO ₂ / kg fuel
MGO	3.93267
ratio	11.2 ×

Why this is the strongest fuel guard in the suite

Epic 3 replaced a single [Co2Factor](#) with a per-engine lookup. A regression that quietly falls back to one constant would be invisible on every MDO/MGO scenario — the two factors are identical. It would also be nearly invisible on 13 (LNG/MGO, 2.753 vs 3.93267, a 30 % gap).

Here it is unmissable.

Verified figures

baseline FOC	8 861.6 t/yr	=	ME 6 880.9	+	AE 1 980.7
baseline CO ₂	10 208 t/yr	=	ME 2 419	+	AE 7 789

Hand-check

ME	:	6 880.9 × 0.35154	=	2 419	✓
AE	:	1 980.7 × 3.93267	=	7 789	✓
sum			=	10 208	✓

Note the inversion: the main engine burns **3.5× more fuel** than the aux side yet emits **3.2× less CO2**. A single-constant model cannot produce that shape at all — if this snapshot ever shows $\text{baselineCO2} \approx \text{totalFOC} \times \text{one factor}$, the per-engine path has regressed.

Fuel price is Ammonia's default **\$1 350/ton**, the most expensive in the table — so the cost panel and the CO2 panel move in opposite directions, which is the real-world trade this feature exists to show.

Takeaway: cheapest $\neq$ cleanest, and the model must keep the two axes independent. This scenario fails loudly if they are ever collapsed.